

WINNER ENGINEERING INTERNAL TRAINING SERIES

MODULE I

INTRODUCTION TO ACMV



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service



CONTENT

1.1 Importance of ACMV System

1.2 Types of Air-Conditioning System

1.3 Types of Mechanical Ventilation System

1.4 Components of ACMV System

1.5 Commonly Used Terms

1.1 IMPORTANCE OF ACMV SYSTEM IN A BUILDING

- To maintain acceptable space temperature
- To maintain acceptable space moisture (relative humidity)
- To maintain room pressure
- To maintain air quality, negate airborne particles, dust and micro-organism
- All and all to provide thermal comfort and acceptable indoor air quality to occupants and processes.

* A well designed ACMV system can provide adequate control for cooling, heating, air filtration for indoor comfort.

1.2 TYPES OF AIR-CONDITIONING SYSTEM

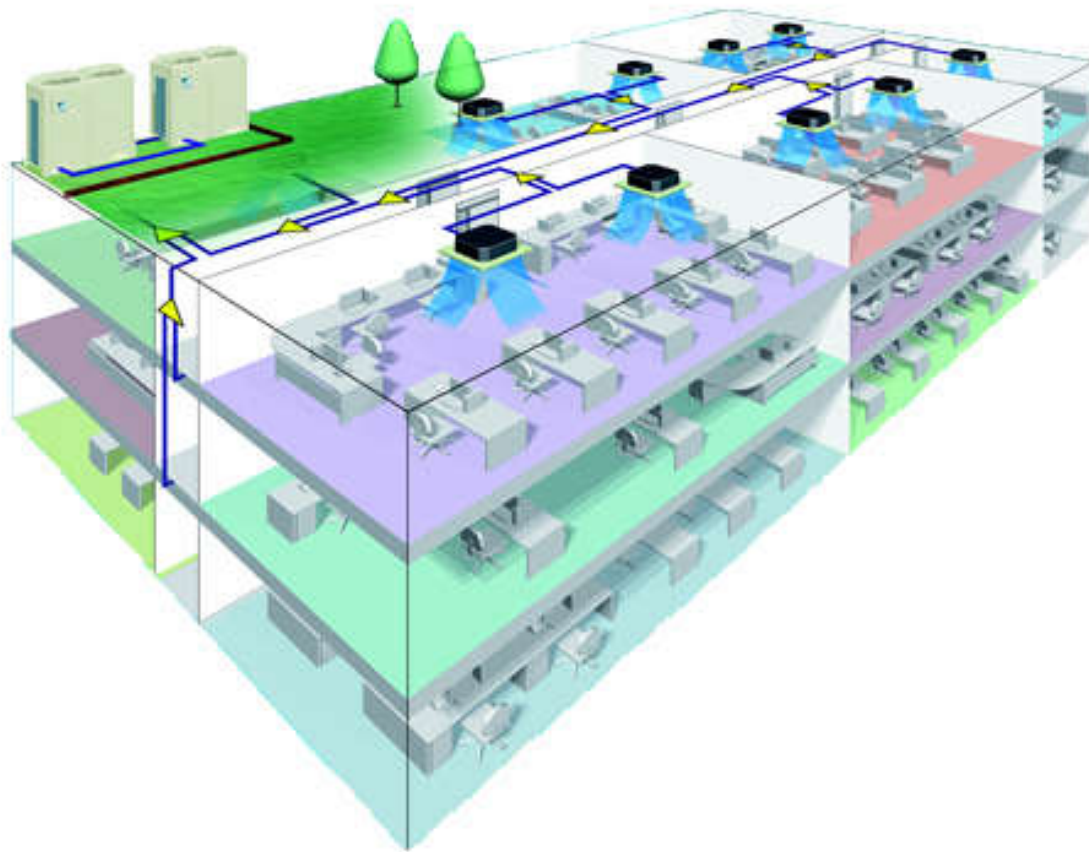
COMMON

- Window AC
- Split unit air-conditioning system (DX-Unit)
 - Single Split
 - Multi Split (VRF or VRV)
- Centralised Air-Conditioning System
 - Water-cooled chilled water system
 - Air-cooled chilled water System

SPECIAL

- DSC (Central District Cooling System)
- Computer room AC (CRAC Unit) System

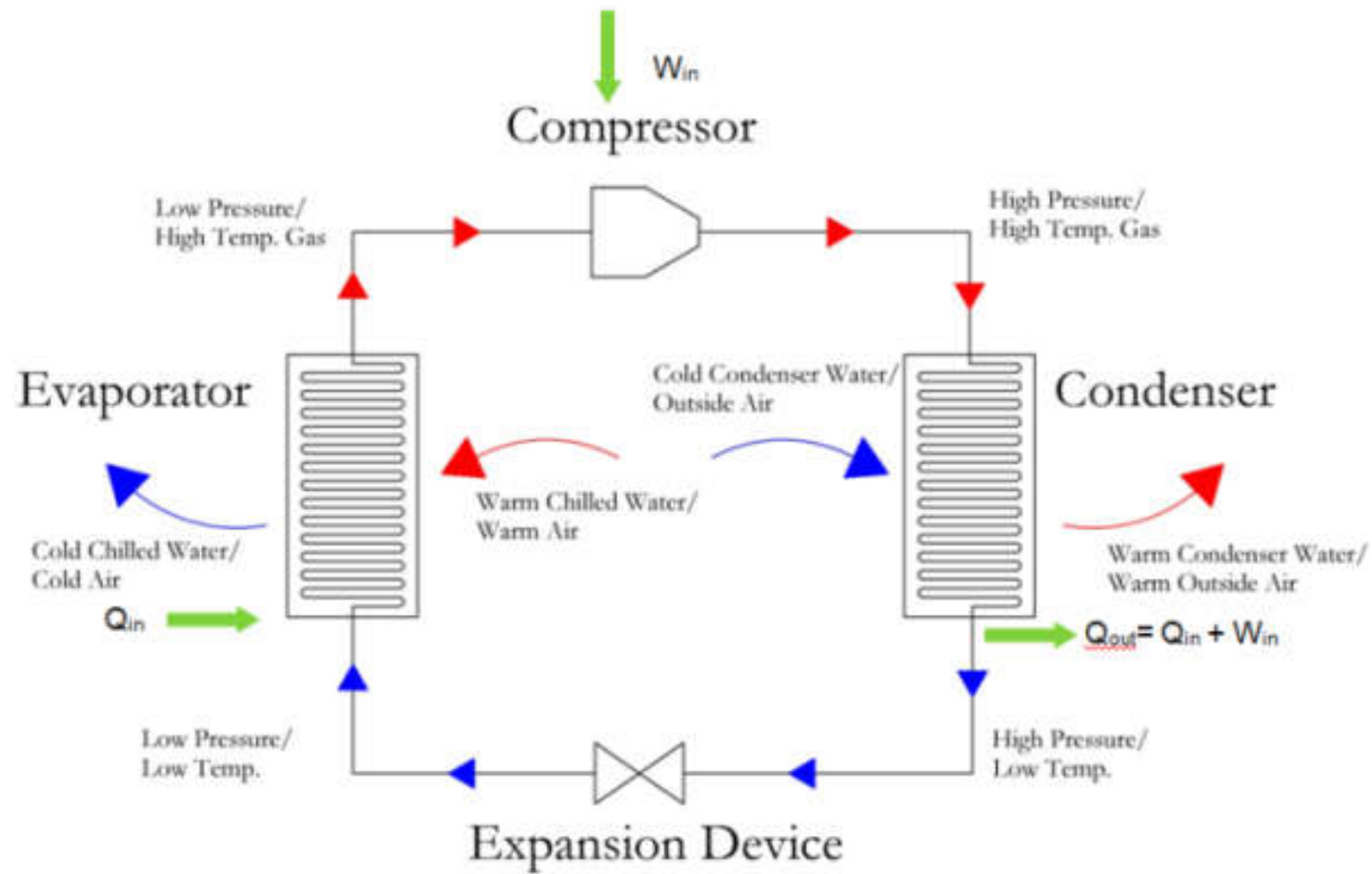




VRF system



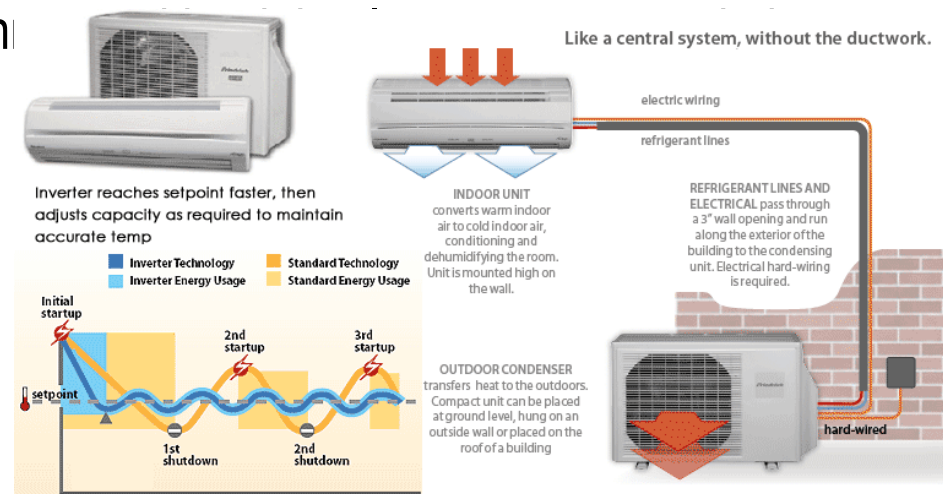
Multi Split system



VAPOUR COMPRESSION REFRIGERANT CYCLE

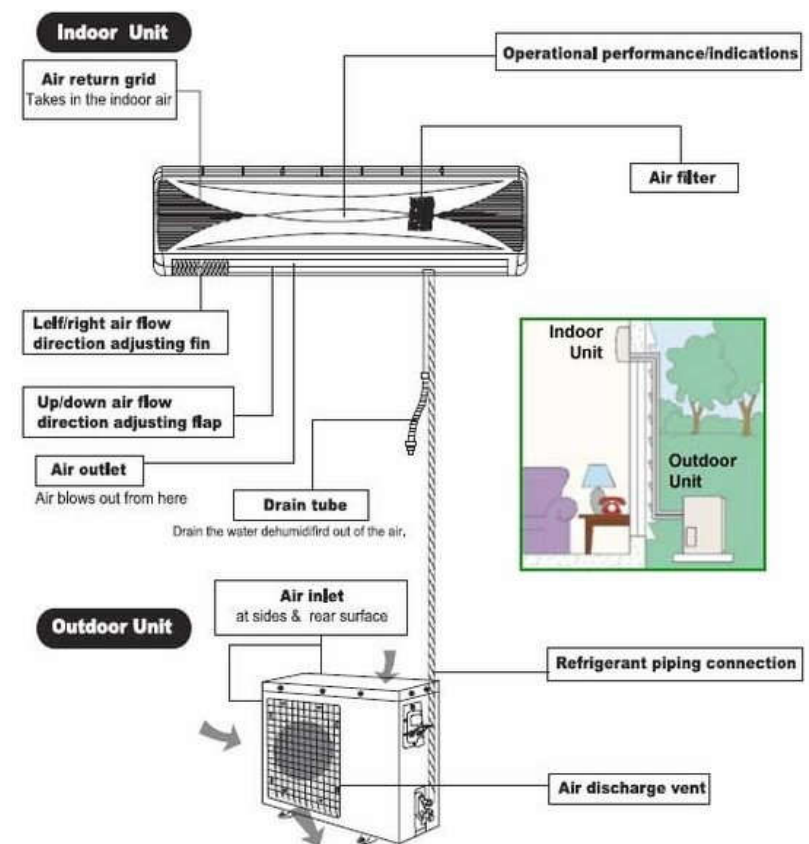
Split Air- Conditioning System

- AKA Direct Expansion (DX) System
- Comprises of 2 parts; outdoor unit (CU) and indoor unit (evaporator)
- Outdoor unit houses components like the compressor, condenser, expansion valve.
- Indoor unit houses the blower, cooling coil.
- Power is supplied to the outdoor unit, which in turn supply to the indoor unit.
- Inverter technology: Adjust compressor frequency according to load to create comfortable environ



Single-Split AC System

- One to one configuration between outdoor unit and indoor unit.
- Power is supplied to the outdoor unit, which in turn supply power to the indoor unit.
- Indoor unit cooling capacity: 2.5 to 7.1kW
- Outdoor unit cooling capacity: Follows indoor u
- Common application
 - Residential bedrooms and family rooms
 - Computer rooms/server rooms
 - Storage rooms
 - Guard houses/ Fire Command Centres



Multi-Split AC System

- One to **many** configuration between outdoor unit and indoor **units**.
- Power is supplied to the outdoor unit, which in turn supply power to the indoor unit. (Depends on capacity)
- Indoor unit cooling capacity: 2.5 to 11.2kW
- Outdoor unit cooling capacity: 5.5 to 20.8kW
- Common application
 - Residential apartmen
 - Small offices

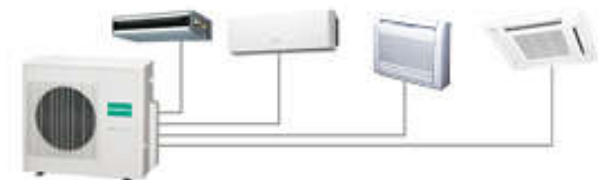
3 Rooms Multi Up to 3 units



2 Rooms Multi Up to 2 units



4 Rooms Multi Up to 4 units



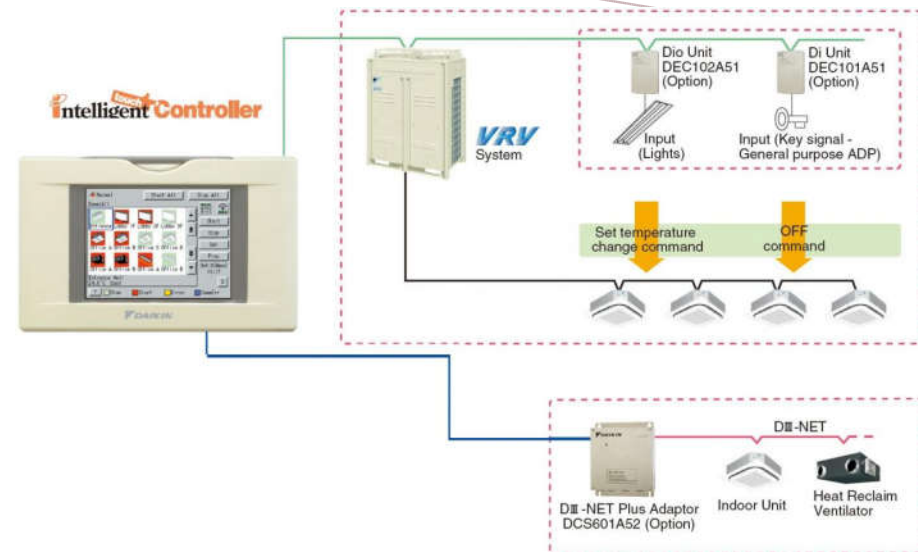
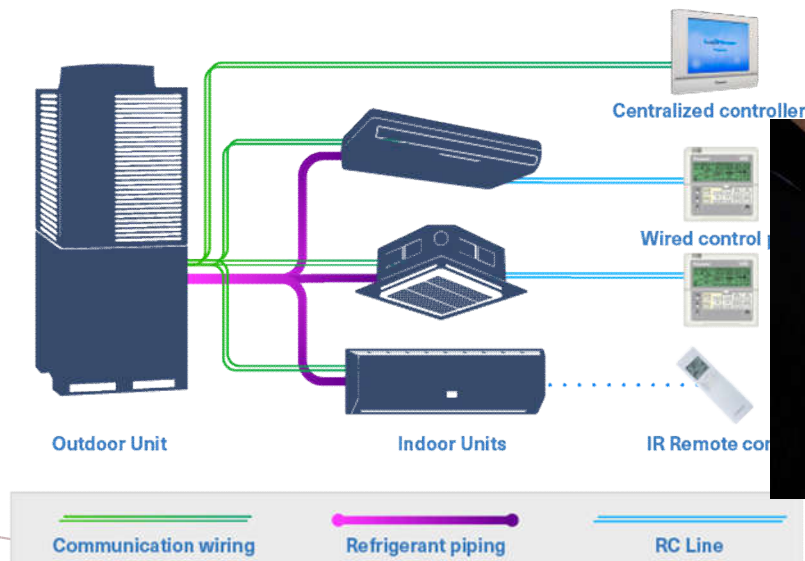
VRF System

- Variable Refrigerant Flow
- Or Variable Refrigerant Volume (Daikin's Trademark)
- Principles:
 - Refrigerant is the only coolant
 - Inverter compressors: lower power consumptions with partial cooling loads
 - Several indoor units can be on the same refrigerant loop/circuit.
 - Ability of modular expansion
- Common applications;
 - Small to medium office buildings
 - Small to medium industrial building
 - Schools
 - Healthcare



VRF System Advantages

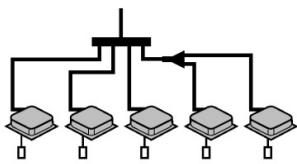
- Centralised control and
- Energy monitoring
- Remote monitoring and control
- Can be integrated to BMS



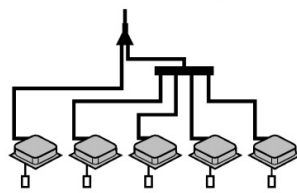
VRF System Advantages

- Long pipe length (up to 165m)
- High max level difference (up to 90m)
- Infinite piping possibilities
- Expandability

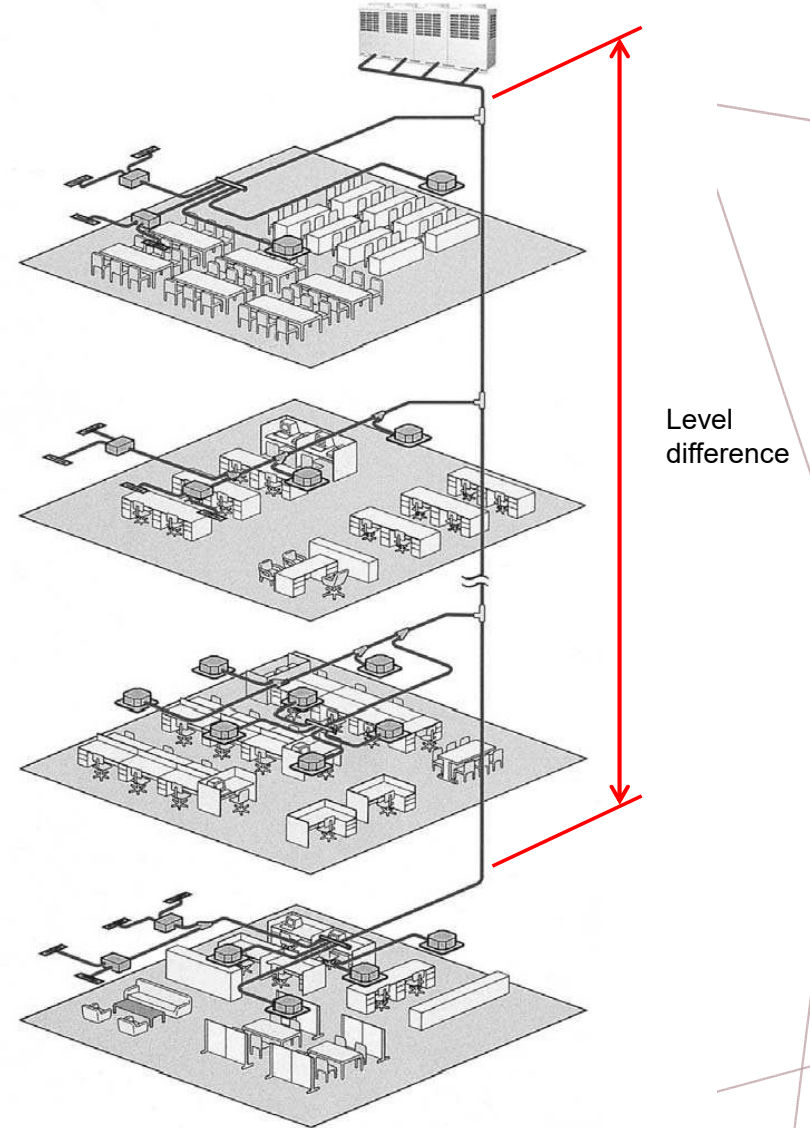
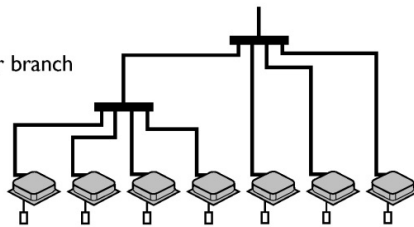
Y joint branch after header branch
(Toshiba unique technology)



Header branch after Y joint

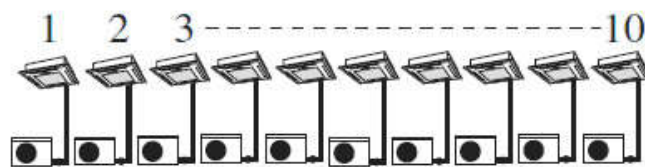
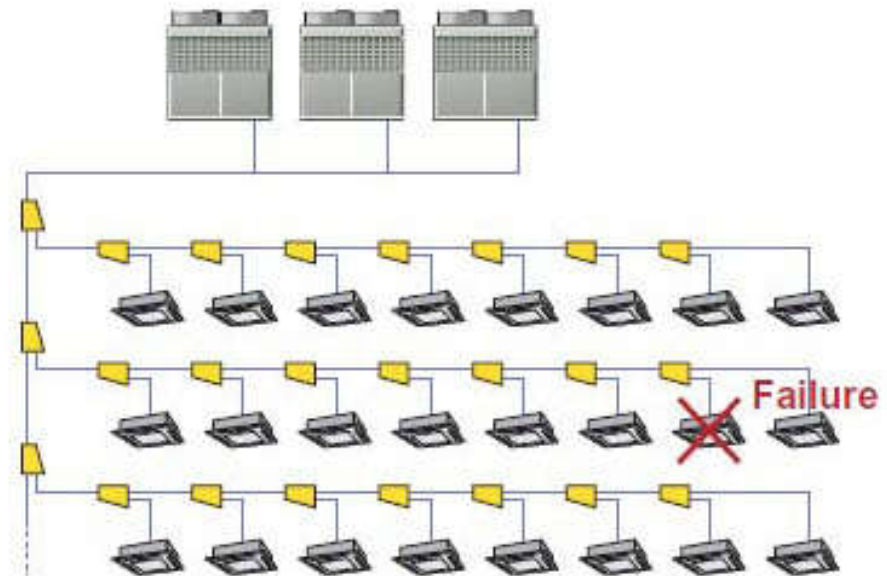


Header branch after header branch

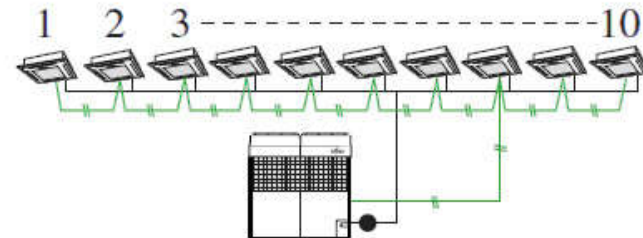


VRF System Advantages

- Space saving, compact
- Less piping
- Other units unaffected by faulty one



Single Split System (10 nos.)

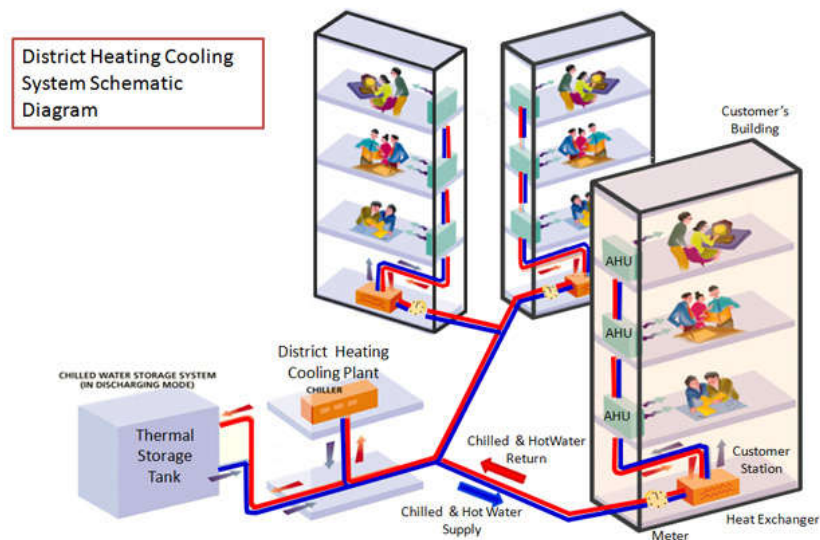


VRF System

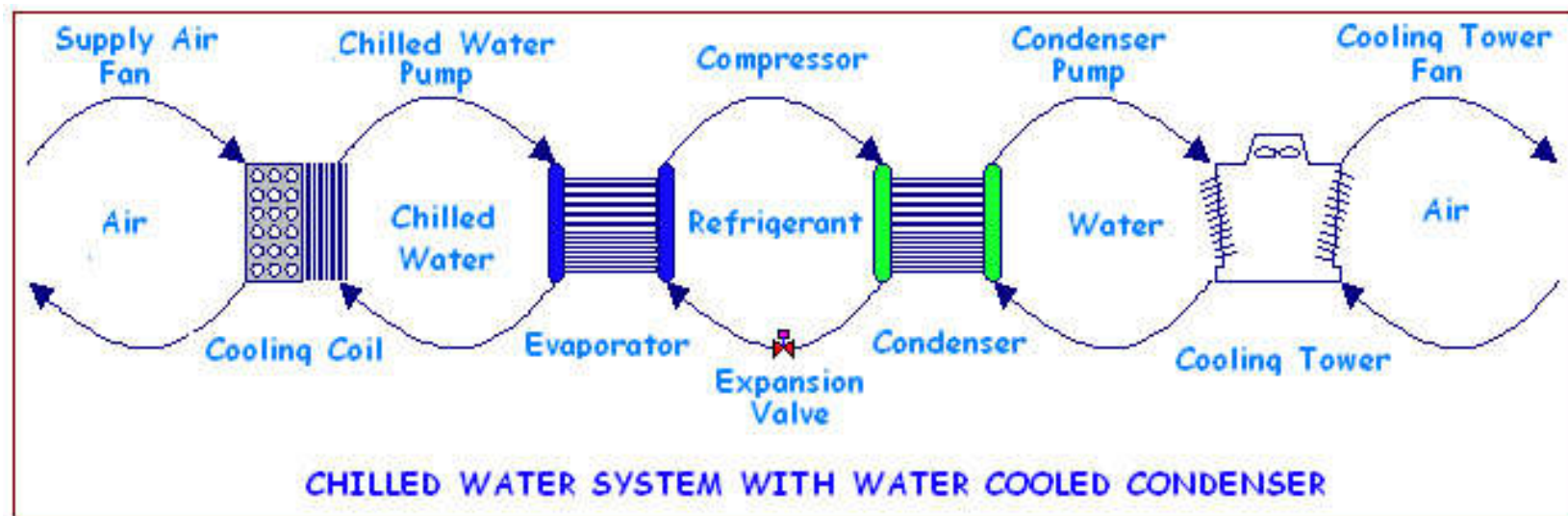
Centralized Air-Conditioning System

- For large buildings, hotels, shopping centres, factories etc that need large cooling capacities.
- Typically uses chilled water to cool and dehumidify the air, instead of refrigerant.

Keppel DHCS District Cooling System

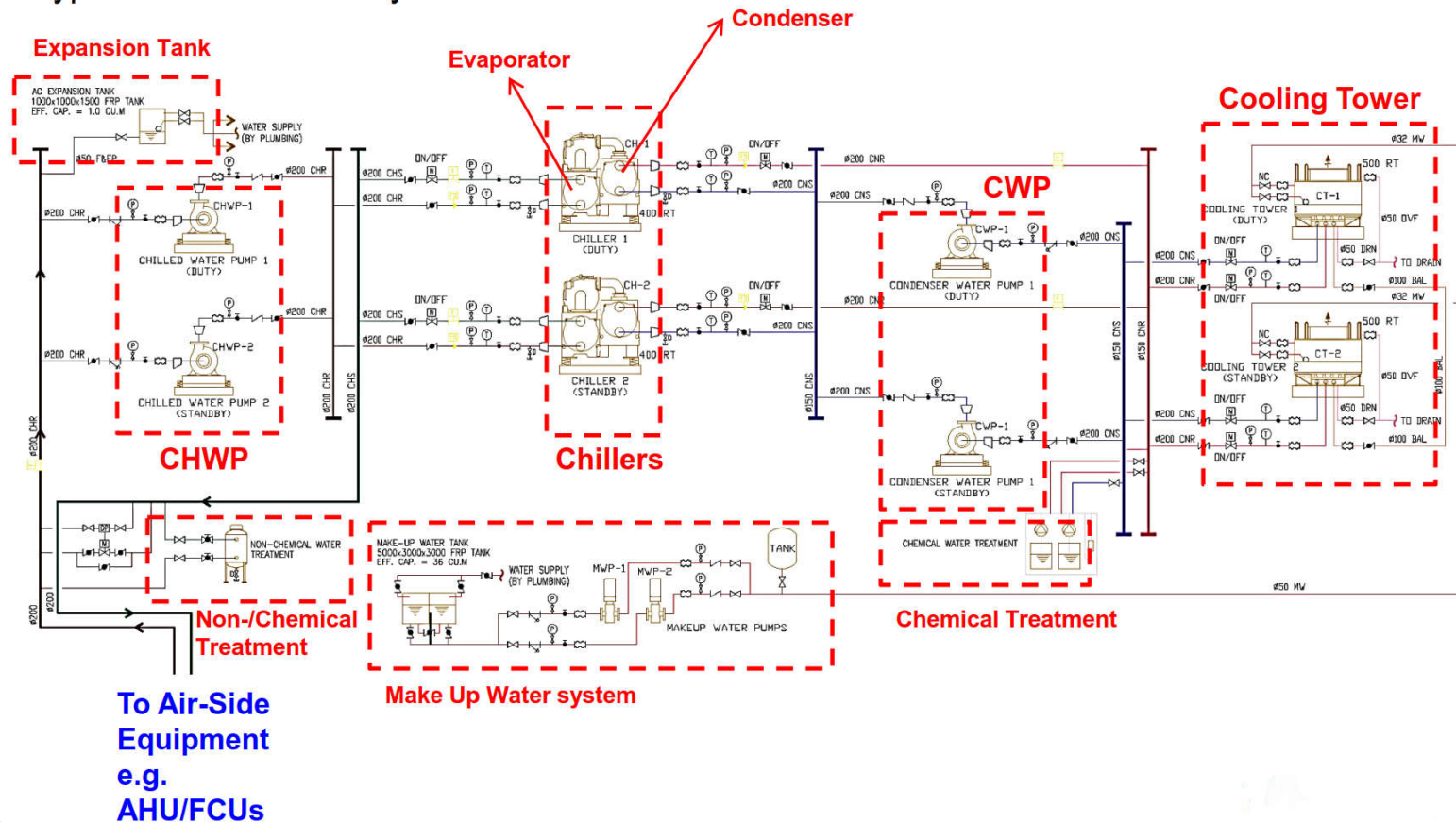


Chilled-Water Loop System



Chilled-Water Loop System

Typical Chilled Water System:



Chilled-Water Loop System

Purpose of the components:

Evaporator (Chiller): Area of the chiller that cools the water

Condenser (Chiller): Area of the chiller that pulls heat off the machine

Cooling Tower: Cools water from the condenser by dissipating it into the air.

Chilled Water Pumps, CHWP: Pumps water from chillers to air-side equipment.

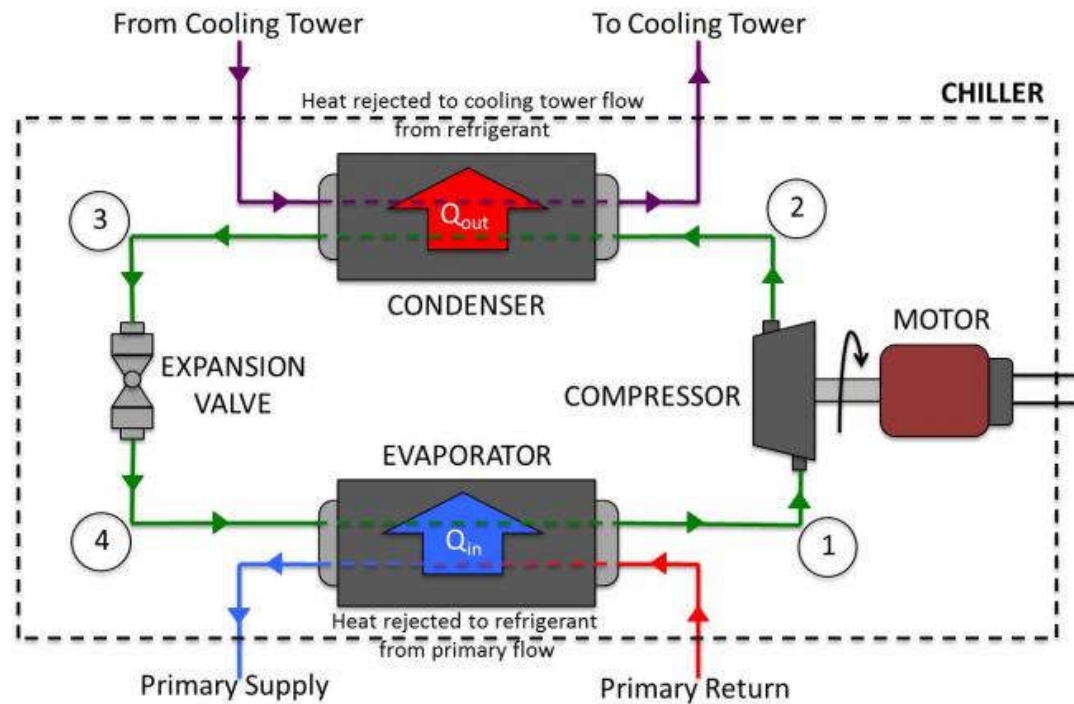
Condenser Water Pumps, CWP: Pumps water from chillers to cooling towers.

Expansion tank: Provides the water for expansion and contraction of water as temperature (and density) changes.

Make up water tank: Replaces water losses due to evaporation, leaks or discharges.

Non-Chemical/Chemical Treatment/Dosing: Injects additives into the system to inhibit corrosion, scaling and build up of contaminants.

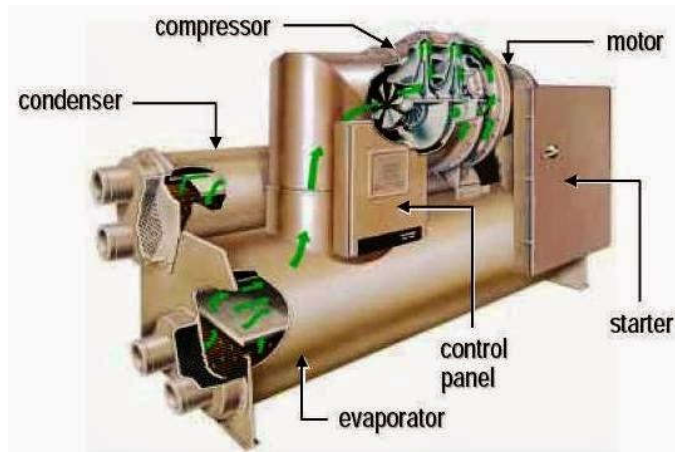
Working Principle – Vapor Compression Cycle



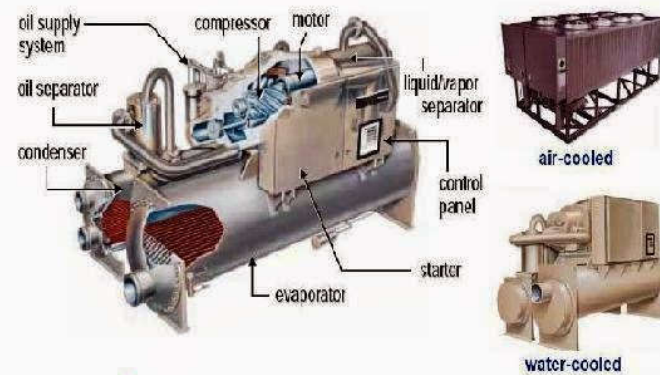
Processes:

- 1 to 2 – Compression of refrigerant vapors in compressor
- 2 to 3 – Heat rejection by refrigerant in condenser
- 3 to 4 – Expansion of refrigerant in expansion valve
- 4 to 1 – Heat addition to refrigerant in evaporator

TYPES OF CHILLERS



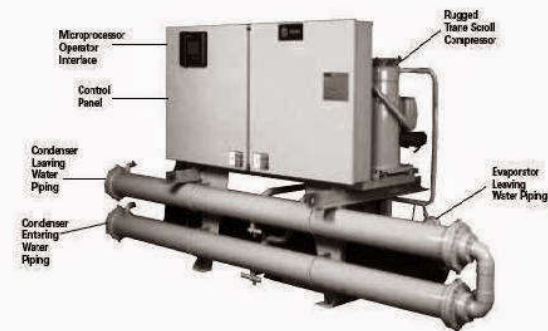
Centrifugal



Screw



Reciprocating



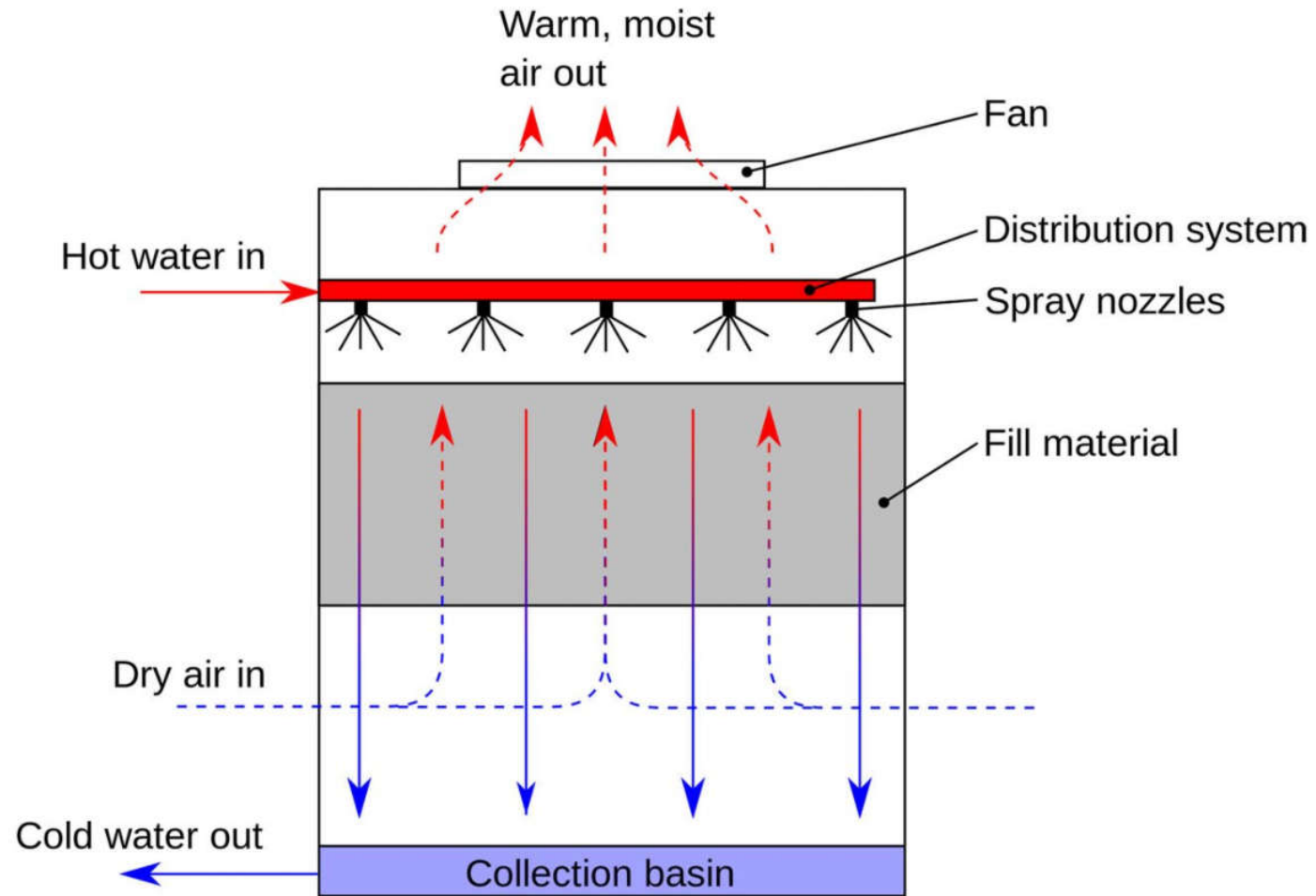
Scroll

TYPES OF COOLING TOWERS

- Crossflow
- Counterflow
- Induced Draft
- Passive Draft



COOLING TOWERS WORKING PRINCIPLE



CHILLED WATER AND CONDENSER WATER PUMPS



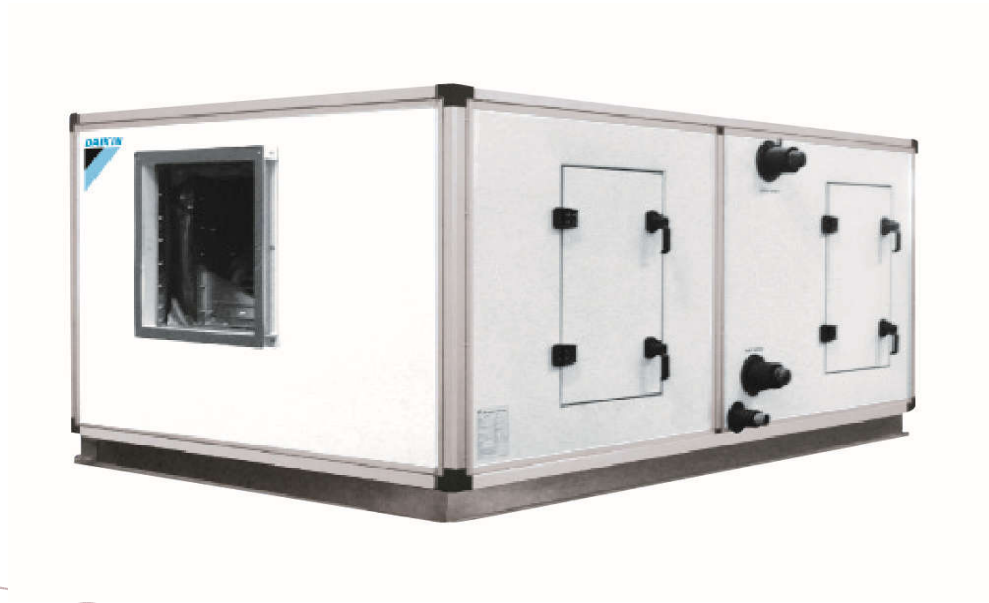
OTHER PARTS OF CHILLED WATER SYSTEM

- Expansion Tank : To cater fluid expansion for a closed loop system (chilled water/evaporator loop side)
- Make-up Water Tank : To top up lost water from evaporation and drift in the cooling tower (Condenser water loop)
- Chemical,non-chemical treatment : Add liquid biocides and chemical that inhibit corrosions and to protect cooling coils of chiller,ahu, and fcu.



Air handling Unit / Fan Coil Unit

To cool, dehumidify, and filter air through a cooling coil and blow air into ductwork distribution system



AHU



FCU

ELECTRICAL PROVISION FOR TYPES OF AC SYSTEM

System Type	Sub-System Type	Power to Fan Coil Units	Power to Condensing Unit
DX Split System	Single Split	No	Isolator
	Multi Split	No	Isolator
	VRV	Power point / Isolator	Isolator
Chilled Water System		Power point / Isolator	NA

1.3 MECHANICAL VENTILATION SYSTEM

Mechanical ventilation is required when natural ventilation is not possible due to the following

- Low level room from the building perimeter.
- Poor local air quality, (eg; next to busy road)
- High local noise level (Unable to open windows)
- Density of occupation, equipment, or high heat load and contamination.
- Control of smell.

TYPES OF MECHANICAL VENTILATION SYSTEM

Pressure System

Fresh air is introduced into the room by fresh air fans which creates a higher internal pressure in relative to outside air.

Vacuum System

Internal air is extracted from the room by exhaust air fan which creates a lower internal pressure in relative to outside air.

Balanced System

Uses both fresh air and exhaust fan to maintain the similar internal pressure to outside

Local Exhaust/Extraction

Extracts local source of heat or contamination. E.g, hoods, fume cupboards, machines.

EXAMPLES OF MECHANICAL VENTILATION SYSTEM

Pressure System

- Fire fighting lobby, smoke stop lobby ventilation
- Staircase Pressurization

Vacuum System

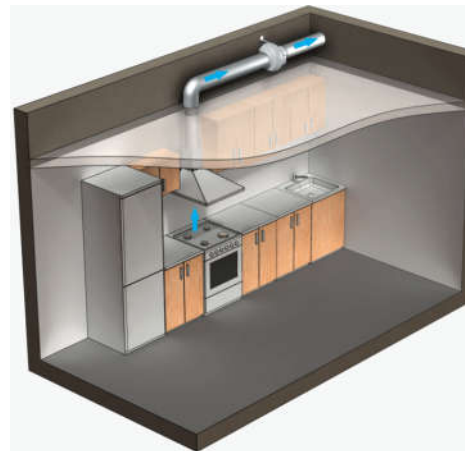
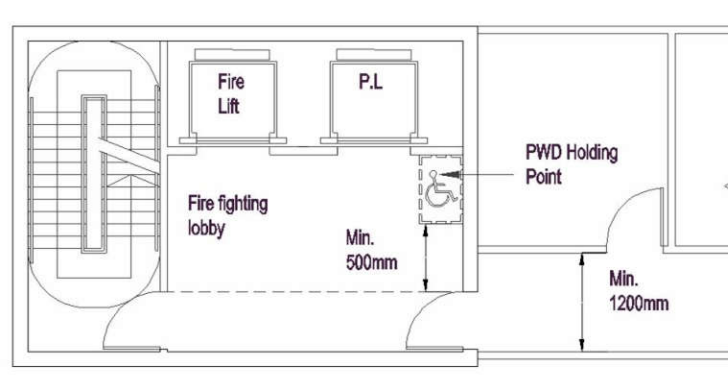
- Toilet exhaust
- Kitchen
- Carpark ventilation

Balanced System

- Smoke purging
- Plantroom

Local Exhaust/Extraction

- Kitchen hood
- Fume cupboard



VACUUM MV SYSTEM

- Draws indoor air by mechanical means out to an outdoor space.
- Sucks air outside the room into the space through cracks and holes.
- Keeps smells and contaminants from spreading into an adjacent room and deposit them outdoor.



Applicable to:

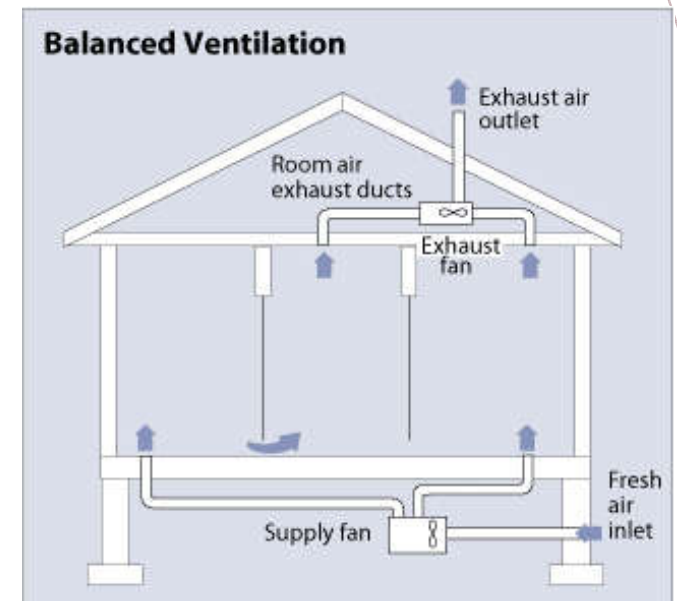
- Toilet exhausts (Keep smells within the toilet)
- Kitchen ventilation (Keep smells within the kitchen)
- Flammable store ventilation (Negative pressure keep smoke from spreading into adjacent rooms)

BALANCED MV SYSTEM

- Pressure of room is usually neutral; amount of air deposited into space is roughly the same as amount of air extracted.
- Most important consideration is exchange or replacement of contaminated/unsafe air in space with outdoor cleaner air.

Applicable to:

- Smoke purging (replace smoke with outdoor air in a fire event)
- Hydrogen purging / CO2 purging (dilutes air concentrated with hydrogen / CO2 to prevent fire and suffocation)
- Plant room ventilation (for rooms that cannot be naturally ventilated and yet not practical to air-condition)



LOCALIZED EXTRACTION

- Pressure of room is usually neutral; amount of air deposited into space is roughly the same as amount of air extracted.
- Most important consideration is exchange or replacement of contaminated/unsafe air in space with outdoor cleaner air.



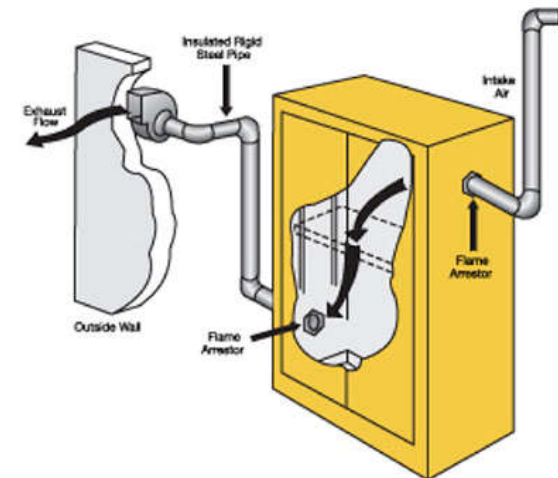
Fume Hood



Extractor Arm

Applicable to:

- Smoke purging (replace smoke with outdoor air in a fire event)
- Hydrogen purging / CO2 purging (dilutes air concentrated with hydrogen / CO2 to prevent fire and suffocation)
- Plant room ventilation (for rooms that cannot be naturally ventilated and yet not practical to air-condition)



Flammable
Cabinet Extract

CIRCULATION FANS

- Only circulates air within the room. No air is supplied nor extracted from the space
- Do not considered as mechanical ventilation in the context of fire safety.



1.4 BREAKDOWN COMPONENT OF ACMV

SYSTEM

- Chiller
- Cooling Tower
- CHWP/CWP
- Expansion Tank
- Make-up Tank
- Chemical Dosing Plant
- Pre-insulated pipework & valves

AIR SIDE

- AHU
- FCU
- VAV
- Venturi Valve
- Duct & Insulation
- Air terminals

REFRIGERANT SIDE

- CU
- REFRIGERANT PIPEWORK

CONTROL

- Field sensors
- DDC/BMS

ELECTRICAL SIDE

- VSD
- Power meter
- Isolators
- Power points

FAN COIL UNITS



4-way Ceiling Cassette FCU



Round Flow Ceiling Cassette FCU



Ceiling Exposed/ Mounted FCU



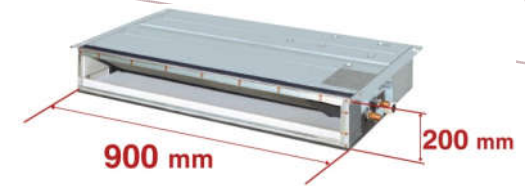
Wall mounted FCU



2-way Ceiling Cassette FCU



Floor Standing FCU



Ceiling Concealed FCU (Slim Type)



Ceiling Concealed FCU (Standard)



Concealed Floor Standing FCU

FANS



In-line/Tube Fan



Bifurcated Fan



Centrifugal Fan



Propeller fan



Direct driven Axial Fan

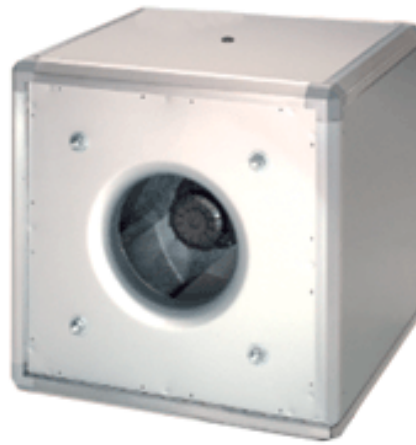


Belt driven Axial Fan

FANS



Roof Fan



Cabinet/Box Fan



Plastic Radial Fan



Roof Ventilator



Jet Fan

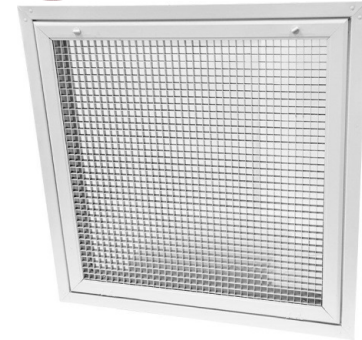
AIR TERMINALS



Linear Diffuser (2 Slot)



Round Supply air grille



Return air grille (Egg crate)



4-way or Square Supply
air diffuser



Bar grille



Double Deflection grille



Louver

EQUIPMENT



VAV Box



Fan Filter Unit



Electric Duct Heater



Electrostatic Air Cleaner

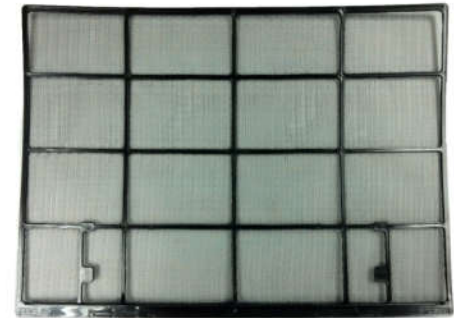
AIR FILTERS



Bag Filter



Box Filter



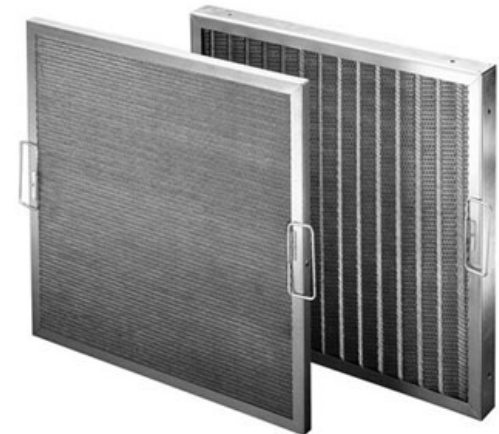
Saranet



HEPA Filter



Pre Filters



Washable Filters

DUCTWORK AND ACCESSORIES



Opposed Blade Damper



Curtain type / Vertical Fire Damper



Butterfly Damper



Bubble tight / Isolation Damper



Motorised Damper



Spring type / Horizontal Fire Damper

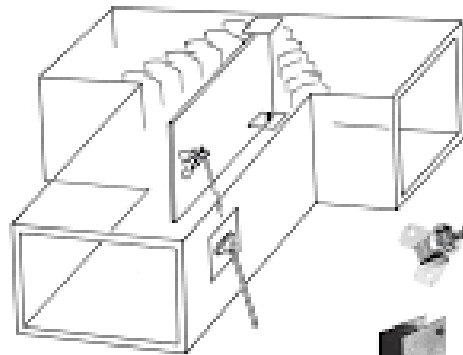


Fusible Link

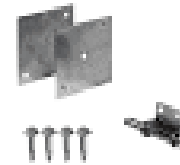
DUCTWORK AND ACCESSORIES



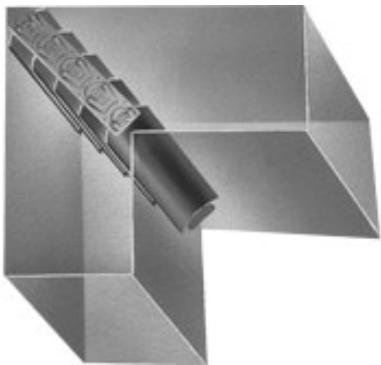
Duct Collars



Splitter Damper



Non-return Damper

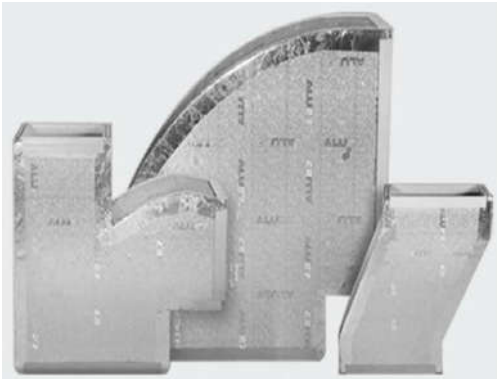


Turning vanes



Pressure Relief Damper

DUCTWORK AND ACCESSORIES



Pre-Insulated Duct (PID)



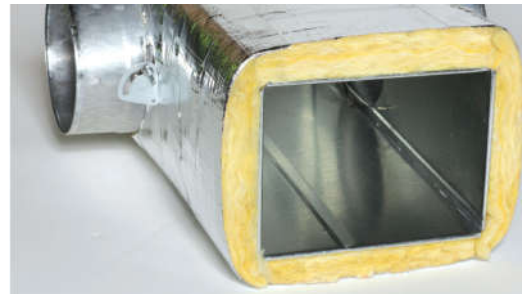
Duct Smoke Detector



Flexible Duct



Galvanised Steel Duct



External duct insulation with
rockwool



Internal Duct Liner
wrapped with
Perforated metal Sheet

ELECTRICAL AND CONTROL



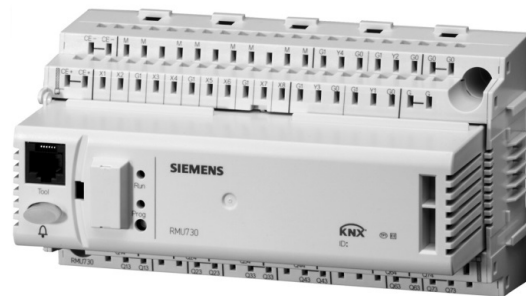
Variable Speed Drive



Mechanical Thermostat



Digital Thermostat



DDC Controller



Temperature Sensor

1.5 COMMONLY USED TERM IN ACMV SYSTEM

Terminology	Description
Air Change	Hourly ventilation rate divided by the volume of a space
Ambient	Normal atmospheric conditions of temperature & pressure
BMS	Building Management System
BTU	Energy required to raise one pound of water one degree Fahrenheit
Coil	Equipment that performs heat transfer to air when mounted inside an air handling unit or ductwork
Condensate	Moisture removed from the air when it is cooled below its dew point; normally associated with moisture removal during the dehumidification process in air conditioning.
Constant air volume (CAV)	System designed to provide a constant air flow, but supply air temperature is varied to meet the thermal loads of a space.
COP	Coefficient of Performance is the ratio of the refrigerant effect (total capacity) versus the work by the compressor over a unit of time; the higher the number the more efficient the system
CHWR	Chilled water return to a chiller
CHWS	Chilled water supply to a cooling unit
CWR	Condenser water return from a chiller
CWS	Condenser water supply from cooling tower

COMMONLY USED TERM IN ACMV SYSTEM

Terminology	Description
Delta T	Difference in temperature of a heating or cooling medium as it enters and as it leaves a system.
Dew Point	The pressure/temperature where refrigerant gas starts changing from vapour to liquid phase. Also used to define the point where air is saturated.
Direct Expansion (DX)	System using a refrigerant that passes through an expansion device prior to entering the evaporator. The system is used to extract heat in the evaporator and reject the waste heat through the condenser.
Dry Bulb Temperature	Temperature of air measured by a thermometer which is freely exposed to the air while it is shielded from radiation and moisture.
EAC	Electronic Air Cleaner, sometimes referred to as ionizers or electronic air purifiers, use electrically charged filters to reduce the number of airborne contaminants
Equivalent Length	The length of straight tubing which has the same pressure drop as the fitting, valve or accessory (of the same nominal size) being considered.
Face Velocity	Measured air speed across an inlet or outlet of a HVAC system
GWP	Global Warming Potential. Measure of global warming relative to CO ₂ =1. CFCs often had GWP between 5,000 and 10,000. HCFCs and HFCs often have GWP between 1,000 and 4,000.

COMMONLY USED TERM IN ACMV SYSTEM

Terminology	Description
HEPA	High Efficiency Particulate Arrestance. Filters that must remove (from air passing through) 99.97% of particles that have a size of 0.3 microns.
Humidity	A measure of the amount of moisture in the air.
Kilowatt (kW)	Measure of energy flow used to express the cooling capacity, heating capacity and power consumption of an air-conditioning system.
Latent Heat	Heat that produces a change in state without a change in temperature.
MERV	Minimum Efficiency Reporting Value. Used to rank the efficiency of air filters. Higher MERV ratings indicate more efficient filter.
N+1	Redundancy that ensures system availability in the event of component s (N) have at least one independent backup component (+1).
OPD	Ozone Depletion Potential. Measure of damage to the ozone layer relative to R11 = 1. CFCs tend to have ODP between 0.1 and 1. HCFCs between 0.01 and 0.1. HFCs, HFOs and Natural Refrigerants = 0.
Pa	Pascal, a measure of pressure unit
Plenum	An enclosed space inside a building or other structure, used for airflow. Often refers to the space between a dropped ceiling and the structural ceiling, or a raised floor and the hard floor.

COMMONLY USED TERM IN ACMV SYSTEM

Terminology	Description
R-22	An HCFC (hydro chlorofluorocarbon) refrigerant often referred to by a brand name such as Freon, used primarily in residential air conditioning systems. It is being phased out for its higher global warming potential (GWP) and ozone depleting properties (ODP).
R-410A	A non-ozone depleting refrigerant that has replaced R-22. Performs at a higher pressure than R-22 and requires the use of different equipment than with R-22.
Recirculating	Ventilation systems that return part of the return air back into the air intake and recirculate the mixture through the room. Little or no air is disposed of to the outside.
Relative Humidity	Ratio of the amount of moisture in the air compared to the amount of moisture the air could hold at the current temperature, expressed as a percentage.
Sensible Heat	Heat that can be measured or felt. Sensible heat always cause a temperature rise.
Single Pass	Systems that take in fresh air from the outside and supply filtered and conditioned air into the room. Whatever volumes of fresh air introduced into the room is ultimately exhausted to the outside.
Thermostat	System that monitors and regulates a heating or cooling system. It can be used to set the desired temperature at which it keeps the environment either heated or cooled.
Ton/Tonnage	Unit of measurement for determining cooling capacity. One ton of cooling is the amount of heat needed to melt a one ton block of ice in a 24 hour period.

COMMONLY USED TERM IN ACMV SYSTEM

Terminology	Description
Two Stage Operation	Two levels of heating or cooling output for enhanced temperature control, energy efficiency and indoor air quality.
VAV	Variable air volume, system that has a stable supply-air temperature, and varies the air flow rate to meet the temperature requirements.
VSD	Variable Speed Drive, equipment that regulates the speed and rotational force, or torque output of an electric motor. Similar to Variable Frequency Drive.
VFD	Variable Frequency Drive, a type of motor controller that drives an electric motor by varying the frequency and voltage supplied to the electric motor.
Ventilation	The intentional introduction of outside air into a space.
Wet Bulb Temperature	Temperature read with a thermometer whose bulb is encased in a wetted wick.
Zones	Sections of a building or space which are controlled independently of each other. Temperature is controlled by different thermostats.